

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination Summer 2024 Course: B. Tech. Semester:III Branch : Electronics and Computer Engineering / Electronics and Computer Science Engineering Subject Code & Name: BTESC304_EC Computer Architecture & Operating System Max Marks: 60 Date: 06/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Give the comparison between computer organization and computer architecture.	1/CO1	6
B)	Explain different types of addressing modes.	2/CO1	6
C)	Explain RISC and CISC architecture.	3/CO1	6
Q.2	Solve Any Two of the following.		12
A)	What are the different issues related in cache memory? How does cache memory works?	2/CO2	6
B)	State the characteristics of Magnetic disk, optical memory and RAID.	3/CO2	6
C)	Explain Principle of operation of paging and Page allocation Hardware support for paging,	4/CO2	6
Q. 3	Solve Any Two of the following.		12
A)	List the applications of Microprogramming.	1/CO3	6
B)	Why does an assembly line in a manufacturing plant refer to as pipelining. Discuss the need of instruction pipe-lining.	3/CO3	6
C)	Explain programmed I/O module and Interrupt driven I/O.	4/CO3	6
Q.4	Solve Any Two of the following.		12
A)	What is operating system ? Give objectives and functions of Operating System (OS).	1/CO1	6

B)	Explain Multiple-Processor Scheduling in detail.	3/CO4	6
C)	What are the advantages of Threads over process . Explain the types of Threads.	4/CO4	6
Q. 5	Solve Any Two of the following.		12
A)	What is the purpose of using semaphores in synchronization ?	4/CO5	6
B)	What is the key factor in preventing deadlocks in a system?	5/CO5	6
C)	What are the different Methods for handling Deadlocks?	2/CO5	6
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